 

# LONG ISLAND MACARTHUR, NY, USA

WMO: **725050**

Lat: **40.794N** Lon: **73.102W** Elev: **84** StdP: **14.65** Time Zone: **-5.00 (NAE)** Period: **94-19** WBAN: **04781**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | 1 | 11.3 | 15.7 | -5.1 | 4.2 | 15.0 | -0.9 | 5.3 | 18.6 | 28.5 | 29.6 | 26.4 | 28.7 | 10.4 | 290 | 0.529 | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | 7 | 14.8 | 88.7 | 73.7 | 85.9 | 72.2 | 83.2 | 71.3 | 77.0 | 83.6 | 75.6 | 81.4 | 74.4 | 79.8 | 10.8 | 220 | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | 75.1 | 132.1 | 80.2 | 73.6 | 125.5 | 78.6 | 72.5 | 120.9 | 77.7 | 40.4 | 83.9 | 39.0 | 81.6 | 37.8 | 79.6 | 83.8 | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | 24.6 | 20.9 | 18.9 | DB | 6.0 | 94.8 | 4.1 | 3.7 | 3.0 | 97.5 | 0.6 | 99.7 | -1.8 | 101.8 | -4.8 | 104.5 | (4) |
| (5) | | | | WB | 4.3 | 79.4 | 3.9 | 2.0 | 1.4 | 80.9 | -0.9 | 82.0 | -3.1 | 83.1 | -5.9 | 84.6 | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | 53.2 | 31.7 | 33.2 | 39.8 | 49.9 | 59.4 | 68.9 | 75.0 | 73.7 | 67.0 | 56.0 | 45.7 | 37.4 | (6) |
| (7) | | | DBStd | 16.72 | 9.70 | 8.53 | 8.07 | 6.92 | 6.52 | 6.15 | 4.79 | 4.76 | 5.88 | 7.26 | 7.85 | 8.41 | (7) |
| (8) | | | HDD50 | 2056 | 569 | 471 | 330 | 83 | 4 | 0 | 0 | 0 | 0 | 24 | 174 | 401 | (8) |
| (9) | | | HDD65 | 5169 | 1031 | 889 | 783 | 455 | 201 | 32 | 1 | 1 | 46 | 293 | 580 | 857 | (9) |
| (10) | | | CDD50 | 3239 | 2 | 1 | 13 | 80 | 294 | 567 | 774 | 734 | 511 | 210 | 44 | 9 | (10) |
| (11) | | | CDD65 | 876 | 0 | 0 | 0 | 2 | 26 | 148 | 310 | 270 | 107 | 13 | 0 | 0 | (11) |
| (12) | | | CDH74 | 5733 | 0 | 0 | 2 | 27 | 185 | 887 | 2311 | 1805 | 471 | 45 | 0 | 0 | (12) |
| (13) | | | CDH80 | 1464 | 0 | 0 | 0 | 7 | 40 | 226 | 679 | 438 | 67 | 7 | 0 | 0 | (13) |
| (14) | Wind | | WSAvg | 9.1 | 10.2 | 10.3 | 10.8 | 10.3 | 8.9 | 8.4 | 8.0 | 7.5 | 8.0 | 8.7 | 9.3 | 9.6 | (14) |
| (15) | Precipitation | | PrecAvg | 46.4 | 3.7 | 3.3 | 4.4 | 4.1 | 3.6 | 3.8 | 3.5 | 4.2 | 3.9 | 4.0 | 3.3 | 4.6 | (15) |
| (16) | | | PrecMax | 59.3 | 8.4 | 6.4 | 9.8 | 7.2 | 7.8 | 8.2 | 6.7 | 10.3 | 8.1 | 14.8 | 5.9 | 8.5 | (16) |
| (17) | | | PrecMin | 37.6 | 1.9 | 1.3 | 0.9 | 1.6 | 0.9 | 0.7 | 1.5 | 0.7 | 1.4 | 0.4 | 1.7 | 1.9 | (17) |
| (18) | | | PrecStd | 5.9 | 1.4 | 1.4 | 2.0 | 1.7 | 1.5 | 2.2 | 1.4 | 2.4 | 1.7 | 2.9 | 1.1 | 1.5 | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | 58.1 | 58.3 | 68.2 | 77.3 | 84.5 | 90.5 | 94.0 | 91.0 | 85.9 | 79.4 | 67.3 | 62.3 | (19) |
| (20) | | | MCWB | | 53.9 | 49.3 | 54.6 | 60.0 | 67.7 | 73.5 | 75.8 | 75.4 | 72.9 | 66.3 | 58.5 | 55.8 | (20) |
| (21) | | 2% | DB | | 53.8 | 52.8 | 59.8 | 69.7 | 79.3 | 85.8 | 89.8 | 87.6 | 81.5 | 73.3 | 63.7 | 57.2 | (21) |
| (22) | | | MCWB | | 51.4 | 47.3 | 49.6 | 56.3 | 64.4 | 71.4 | 74.2 | 73.5 | 69.6 | 66.2 | 58.3 | 54.2 | (22) |
| (23) | | 5% | DB | | 50.1 | 49.5 | 54.9 | 64.5 | 74.4 | 82.0 | 86.4 | 84.4 | 79.0 | 70.5 | 61.3 | 54.1 | (23) |
| (24) | | | MCWB | | 46.8 | 44.9 | 48.0 | 52.5 | 62.6 | 69.4 | 72.8 | 72.1 | 68.8 | 64.1 | 57.2 | 51.5 | (24) |
| (25) | | 10% | DB | | 46.1 | 46.0 | 51.6 | 61.0 | 70.5 | 79.2 | 83.6 | 82.0 | 76.5 | 67.7 | 58.5 | 50.9 | (25) |
| (26) | | | MCWB | | 42.6 | 41.8 | 45.8 | 51.1 | 60.4 | 67.9 | 72.0 | 71.1 | 68.2 | 62.1 | 54.1 | 47.2 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | 55.3 | 53.1 | 58.0 | 63.6 | 70.9 | 76.1 | 79.5 | 78.9 | 76.0 | 71.6 | 63.3 | 58.5 | (27) |
| (28) | | | MCDB | | 57.1 | 54.7 | 66.1 | 75.1 | 81.2 | 86.0 | 89.4 | 87.2 | 80.9 | 73.8 | 64.9 | 60.5 | (28) |
| (29) | | 2% | WB | | 51.8 | 49.3 | 52.7 | 59.3 | 67.8 | 73.8 | 77.2 | 77.2 | 74.0 | 68.6 | 60.6 | 54.8 | (29) |
| (30) | | | MCDB | | 53.1 | 51.5 | 56.4 | 65.4 | 75.3 | 81.7 | 84.6 | 82.6 | 77.6 | 72.1 | 62.5 | 56.3 | (30) |
| (31) | | 5% | WB | | 47.5 | 45.9 | 49.7 | 56.2 | 65.5 | 72.1 | 75.7 | 75.7 | 72.5 | 66.0 | 57.9 | 51.9 | (31) |
| (32) | | | MCDB | | 49.4 | 48.0 | 53.5 | 61.0 | 71.9 | 78.6 | 82.0 | 80.7 | 76.0 | 69.1 | 60.4 | 54.0 | (32) |
| (33) | | 10% | WB | | 43.4 | 42.0 | 46.4 | 53.7 | 62.8 | 70.4 | 74.5 | 74.3 | 70.7 | 63.4 | 55.1 | 47.8 | (33) |
| (34) | | | MCDB | | 45.7 | 45.7 | 50.2 | 57.9 | 68.3 | 76.1 | 80.4 | 79.0 | 74.5 | 66.5 | 57.9 | 50.3 | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | 13.8 | 14.7 | 15.1 | 16.5 | 16.5 | 15.4 | 14.8 | 14.6 | 15.3 | 16.2 | 15.2 | 13.7 | (35) |
| (36) | | 5% DB | MCDBR | | 17.0 | 18.0 | 19.8 | 22.1 | 22.0 | 19.1 | 17.6 | 16.5 | 16.4 | 17.9 | 16.4 | 16.2 | (36) |
| (37) | | | MCWBR | | 15.5 | 14.4 | 13.1 | 11.8 | 11.2 | 8.9 | 7.1 | 7.4 | 8.6 | 11.5 | 13.3 | 15.3 | (37) |
| (38) | | 5% WB | MCDBR | | 16.4 | 17.2 | 18.0 | 17.5 | 18.7 | 15.9 | 14.4 | 13.0 | 12.5 | 14.8 | 14.3 | 15.2 | (38) |
| (39) | | | MCWBR | | 16.4 | 16.3 | 14.3 | 12.0 | 10.7 | 8.7 | 7.1 | 6.8 | 8.2 | 11.8 | 13.8 | 16.0 | (39) |
| (40) | Clear-Sky Solar Irradiance | | taub | | 0.307 | 0.314 | 0.336 | 0.381 | 0.431 | 0.476 | 0.488 | 0.467 | 0.396 | 0.345 | 0.317 | 0.307 | (40) |
| (41) | | | taud | | 2.522 | 2.502 | 2.483 | 2.378 | 2.272 | 2.185 | 2.162 | 2.215 | 2.385 | 2.540 | 2.567 | 2.567 | (41) |
| (42) | | | Ebn at Noon | | 272 | 287 | 291 | 283 | 270 | 257 | 252 | 254 | 267 | 272 | 266 | 264 | (42) |
| (43) | | | Edh at Noon | | 23 | 27 | 31 | 37 | 42 | 46 | 47 | 43 | 34 | 26 | 22 | 21 | (43) |
| (44) | All-Sky Solar Radiation | | RadAvg | | 585 | 868 | 1199 | 1543 | 1723 | 1886 | 1910 | 1710 | 1394 | 959 | 664 | 499 | (44) |
| (45) | | | RadStd | | 49 | 84 | 101 | 156 | 170 | 180 | 98 | 108 | 112 | 90 | 55 | 44 | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD50 | HDD65 | CDD50 | CDD65 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | +144 | +86 | (46) |
| (47) | Regional (69 neighbors) | +0.72 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | +202 | +113 | (47) |

Nomenclature: See separate page